Final Project Report

Network Intrusion Detection System (NIDS)

1. Dataset Description

The project utilizes the NSL-KDD dataset, a refined version of the KDD'99 dataset.

- Records: 148,517 total records.
- Features: 41 original attributes mapped to 122 features after encoding.
- Target Categories: Normal, DoS, Probe, R2L, and U2R.

2. Preprocessing & Feature Engineering

- Data Cleaning: Stripped periods from labels and dropped the non-predictive 'difficulty' column.
- Categorical Mapping: Hierarchically grouped specific attacks into 5 major categories.
- Encoding: One-Hot Encoding for categorical features; Label Encoding for the target.
- Scaling: StandardScaler applied to ensure uniform feature influence.
- Handling Imbalance: Implemented SMOTE with a custom strategy to oversample minority classes (U2R and R2L) to a minimum of 5,000 samples.

3. Data Visualization

- Class Distribution: Identified the dominance of Normal and DoS traffic.
- Confusion Matrices: Verified model performance across all categories.
- Feature Importance: Identified src_bytes and count as the most influential predictors for the Decision Tree.

4. Model Implementation and Evaluation

- Model 1: Decision Tree Classifier
    - Accuracy: 99.43%
    - Strengths: Exceptional speed and the highest precision for the R2L category.
- Model 2: ANN (MLPClassifier)
    - Architecture: 2 Hidden Layers (128, 64 nodes).
    - Accuracy: 98.99%
    - Strengths: Robust generalized learning with non-linear pattern recognition.

5. Model Comparison and Conclusion

Metric	Decision Tree	ANN (MLP)
Overall Accuracy	99.43%	98.99%
R2L F1-Score	0.93	0.89
U2R F1-Score	0.65	0.62
Training Speed	Fast	Moderate

Conclusion: While both models exceeded 98% accuracy, the Decision Tree Classifier is recommended for this NIDS implementation. It provided superior detection for the high-risk R2L and U2R minority classes while maintaining lower computational overhead.